Shadow-Enhanced Hadamard Test

Team 8 — Garbage Collectors · Qiskit Hackathon 2026 · Topic 5

checkpoints 56/56 pass | results ledger 59 sourced rows | IBM QPUs 3 devices, 17 jobs | qiskit 2.5 | license MIT

Every Hadamard test throws away half its data. **We recycle it.**

Quickstart · Results · Gallery · Repo map · Honesty · Slides

A standard Hadamard test measures one ancilla qubit and **throws the system register away**. This project keeps it. By measuring those discarded qubits in randomly chosen bases they become a *classical shadow*, so a single set of shots yields not just the interference signal $\chi(t)$ but the system's observables and — the point of the whole exercise — a **symmetry label for every recovered energy level**. We implement the protocol of Faehrmann, Eisert & Kueng (PRL **135**, 150603, 2025) end to end on a 3-qubit XXZ benchmark, validate every estimator against exact references, quantify what the extra information costs, and characterise where the method breaks — on simulators and on three real IBM devices.

Every number in this repository traces to the command that produced it (see `RESULTS.md` and `EVIDENCE.md`). That includes the numbers we got wrong and retracted.

Symmetry-resolved spectrum — the colour is the deliverable

The deliverable in one picture: every recovered energy level carries a symmetry label extracted from the register everyone else throws away.

- ▶ **The idea in one figure** — what a classical shadow actually is
- ▶ **The benchmark** — 3 spins, nearest-neighbour only, exactly solvable

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5-minute quickstart (reproduces the headline numbers from saved data)

No long reruns, no quantum hardware, no IBM account needed.

```
git clone https://github.com/mnsh0409/Qiskit-Hackathon-2026.git
cd Qiskit-Hackathon-2026
python -m venv .venv && source .venv/bin/activate
pip install "qiskit==2.5.1" "qiskit-aer==0.17.2" "numpy<2.5" "scipy<1.18" \
            matplotlib pylatexenc

# the headline result, read back from the saved 12-seed summary in data/
# (the per-seed CSVs behind it are also in data/; regenerate with
evidence/scripts/seed_sweep.py)
python -c "
import json; d = json.load(open('data/multiseed_summary.json'))
print(f\"{'exact E':>9} {'recovered':>20} {'exact q':>8} {'recovered q':>16}  seeds OK\")
for r in d['rows']:
    print(f\"{r['E_exact']:+9.4f} {r['E_mean']:+11.4f} +- {r['E_sd']:.4f} \"
          f\"{r['q_exact']:+8d} {r['q_mean']:+11.3f} +- {r['q_sd']:.3f}
{r['labels_ok']}/{r['n_seeds']}\")
"
```

Expected output: four energy levels with across-seed sd 0.002-0.013 (max deviation from exact 0.004), and **12/12 seeds label every level correctly**.

Then open `figures/07_symmetry_resolved_spectrum.png` — the money plot. Grey is the plain windowed Fourier transform of the ancilla signal; the coloured stems are our reconstruction, coloured by the symmetry sector recovered from the recycled data. (The grey blur is an *estimator* limitation, not a limitation of the standard protocol — R021 shows the standard protocol with a good estimator recovers energies just as well. What it cannot do, at any shot count, is produce the colour.)

Full rerun (~10 min, all 56 checkpoints, regenerates every figure):

```
pip install "ipykernel>=6.29,<7" nbconvert==7.17.1 nbclient==0.11.0
python -m ipykernel install --user --name qh26-t5
jupyter nbconvert --to notebook --execute --inplace \
  --ExecutePreprocessor.kernel_name=qh26-t5 shadow_hadamard_challenge_PARTICIPANT.ipynb
```

> **Environment note (real, cost us time):** `requirements.txt` pins `numpy==2.5.2` / `scipy==1.18.0`, which need **Python 3.12**. On 3.11 the install fails outright. The relaxed pins above are verified-good on 3.11 with all 56 checkpoints passing (logged as BUGLOG B01).

Figure gallery

figure	what it shows
07_symmetry_resolved_spectrum.png	**Money plot** — recovered spectrum, colour = symmetry sector from the recycled register
06_fundamental_validation.png	Four-panel validation: χ quadratures, joint observable, N^(−1/2) scaling
06_conservation.png	Conserved ⟨Q⟩ flat in time vs non-conserved $\langle Z_0 \rangle$ drifting — the pooling rule, shown
07_chi_vs_chiQ_spectra.png	χ vs χ_Q side by side — the ratio at each peak *is* the quantum number
09_noise_comparison.png	Noise damps amplitude; peak positions survive, weights do not
08_krylov_regularisation.png	Krylov GEVP instability — an honest look at an ill-conditioned method
03_circuit_phi_0_real.png	The circuit, real quadrature

Track B, AQC and the method boundaries (generated by `evidence/scripts/make_*_chart*.py`):

figure	what it shows
fig_aqc_scaling.png	AQC crossover at n=4, 36× by n=7 — and the phase trap, with its one-gate fix
fig_tb_anticontrol.png	What anti-control *is*: the X–cW–X sandwich that puts two dynamics on one ancilla
fig_tb_baselines.png	Verification cost vs capability against the Loschmidt echo and Hilbert–Schmidt test
fig_tb_hardware.png	Two QPUs, four arms, one job — including where the cheap baseline beats us
fig_tb_symmetry.png	A free error bar: ⟨Q⟩_W − ⟨Q⟩_U must be exactly zero, so its deviation is pure device error
fig_tb_dos.png	The same circuit as a density-of-states estimator (Goh & Koczor Eq. 7)
fig_skqd_boundary.png	Where SKQD helps and why not here — the Δ/J localisation boundary, 1D vs 2D
fig_model.png	The benchmark Hamiltonian: 3 spins, nearest-neighbour only
fig_shadow.png	What a classical shadow is, worked through one record

figure	what it shows
fig_ablation.png, fig_escalation.png, fig_direct_z.png	Protocol × estimator ablations; the four ways to treat the garbage

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What's here

path	what it is
`shadow_hadamard_challenge_PARTICIPANT.ipynb`	The graded notebook — **56/56 checkpoints pass**, outputs embedded
`RESULTS.md`	Every reported number, with its producing command
`EVIDENCE.md`	Row-to-artifact index — how to verify each result
`BUGLOG.md`	Bugs found, root causes, prevention rules — including our own retractions
`CONVENTIONS.md`	Frozen conventions (operators, signs, endianness, shot budget)
`EXPLAINER.md`	Plain-language description for non-specialists
`hardware_run.py`	Submit/fetch hardware jobs on any IBM backend (3 commands)
`layout_search.py`	Pick the best-calibrated physical qubit window before submitting
`HANDOVER.md`	Instructions for collaborators with hardware we can't reach
`evidence/`, `data/`, `real_machine/`	Producing scripts, per-seed CSVs, raw hardware job dumps
`deck/`	Slides + speaker script

Headline results

AQC makes the Hadamard test scale — and a trap that would have broken it

A Hadamard test's controlled evolution is its bottleneck: the standard exact block costs ~4^(n+1) two-qubit gates. Approximate Quantum Compiling replaces it, with the crossover measured at n=4 and **36× fewer gates by n=7** — *as a compilation result*. **We then ran it on two QPUs and it did not translate**: at n=6 every arm on both `ibm_marrakesh` and `ibm_kingston` returned noise — the 576-gate AQC arm at 0.026 and 0.033 survival against the 0.315 / 0.401 we recorded before submitting. That result is on the slides as the headline it is. [R054] But the compiler optimises *state fidelity*, which is blind to global phase — and a Hadamard test is an interferometer that measures exactly that phase. Shipping it naively returns a χ wrong by 2.3–3.0 radians, an error as large as the signal. **One ancilla phase gate fixes it**: |Δχ| 1.33 → 0.008.

AQC scaling and the phase trap

Where a cheaper baseline beats us, on two QPUs

We benchmarked our Track B arm against the methods the compiling literature actually uses. The 2-gate Loschmidt echo is cheaper *and*, on hardware, 27× more accurate. We lead with that rather than bury it.

Track B on hardware, two devices

- **56/56 checkpoints pass** on the graded notebook, zero hard failures.

- **All four energy levels** recovered with **every symmetry label correct**, reproducibly across **12/12 independent seeds**. [R012, R013, R019]
- **AQC makes the controlled evolution scale — on paper**: crossover at n=4, **36× fewer two-qubit gates at n=7**, surviving a 2D lattice, Fermi–Hubbard, and heavy-hex routing (which in fact *widens* the gap). [R046, R049, R051, R052]
- **…and we tested whether that reaches hardware. On two QPUs, it does not.** At n=6 every arm on marrakesh *and* kingston returned noise (AQC 0.026 / 0.033 vs pre-registered 0.315 / 0.401). On both machines the exact-block arm's apparent "survival" (**1.485**, **1.191**) is *unphysical* — a T_1 relaxation floor that a magnitude-only metric scores as the **winner**. [R054]
- **A phase trap found and fixed** for the cost of one single-qubit gate — a phase-blind compiler feeding a phase interferometer. [R046]
- **Honest cost accounting:** the shadow route replaces **13 dedicated experiments** at a measured **1.07× variance premium** — and the 3^w variance model predicts the observed error bars to within 2%. [R021, R029]
- **Real hardware, both current IBM architectures** — heavy-hex Heron *and* a Nighthawk-class square-lattice device (`ibm_miami`, 218 couplers; identified from its own coupling map in `real_machine/log.txt`). Signal survival **0.878** on the frozen benchmark once the circuit is shallow enough, and a genuine random-basis shadow ensemble recovering a valid ⟨Q⟩ at **0.957 survival**. Depth beats device choice by 2.4–4.6× vs 1.1–2.1×. [R018, R023]
- **A self-validating decoherence witness** that needs no exact diagonalisation, so it remains usable where classical verification is impossible. [R033]

Reproducibility and honesty commitments

- All Part A/B results derive from **one shared record set** — 128 settings × 2000 shots = **256,000 shots**, seed 2026. Side studies (12-seed reproducibility, scaling, noise, ablations, hardware) are declared separately and never mixed into headline numbers.
- Hardware results support **robustness claims only** — never a headline number.
- Statistical claims rest on ≥10 seeds with bootstrap CIs; the bootstrap itself was validated against real seed-to-seed spread. [R019, R020]
- We make **no claims of outperforming classical computation.** The system is 3 qubits; a laptop diagonalises it instantly. It is small *on purpose*, so every estimate can be graded against an exact answer.
- Retracted numbers stay visible with their corrections (see R008 and BUGLOG B04).

Citing

If this is useful to you, please cite the underlying method:

> P. Faehrmann, J. Eisert, R. Kueng, *In the Shadow of the Hadamard Test*, Phys. Rev. Lett. **135**, 150603 (2025). arXiv:2505.15913

and link back to this repository for the implementation and characterisation.

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